

CS201 Midterm 1 Part B

13 March 2025

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Abstract—Write only ONE function per question. Do not check the input, assume that the input is correct. Unless noted, (i) the input data structure is not empty, (ii) time complexity of the algorithm does not matter, (iii) you can not use extra data structures including arrays, (iv) you can use any function given in the data structure, (v) you can not modify the data structure contents.

I. QUESTION (LINKEDLIST) (14 POINTS)

In math, any number can be represented as the sum of distinct Fibonacci numbers. For example, given the first 10 Fibonacci numbers

1 2 3 5 8 13 21 34 55 89

The number 100 can be represented as $89 + 8 + 3$. In this question, you will identify the Fibonacci representation of a number N , which works as follows.

- Creates a temporary linked list of Fibonacci numbers in reverse until N . So, for the given example, the list will contains the numbers 89, 55, 34, 21, 13, 8, 5, 3, 2, 1.
- Using Fibonacci numbers list, prints the representation with spaces in between.

You are not allowed to use any linked list methods.

```
void printFibonacciWay(int N)
```

For 72, the output will be

55 13 3 1

For 100, the output will be

89 8 3

For 89, the output will be

89

For 1, the output will be

1

II. QUESTION (DOUBLYLINKEDLIST) (16 POINTS)

Write the method

```
DoubleLinkedList[] divideList(int k)
```

which constructs an array of linked lists by dividing the original linked list into k equal parts. The first, second, ..., k 'th element of the original linked list will be the first element of the first, second, . . . , k 'th output linked list, etc. The elements of the output linked list should be created (not copied from the original linked list). You are not allowed to use any linked list methods.

For the linked list of

3 5 2 1 6 7 4 3 2 1 5

with $k = 4$, the output linked lists will be

```
3 6 2
5 7 1
2 4 5
1 3
```

For the linked list of

3 5 2 1 6 7 4 3 2 1 5

with $k = 3$, the output linked lists will be

```
3 1 4 1
5 6 3 5
2 7 2
```

For the linked list of

3 5 2 1 6 7 4 3 2 1 5

with $k = 2$, the output linked lists will be

```
3 2 6 4 2 5
5 1 7 3 1
```

III. QUESTION (STACK) (10 POINTS)

Write the method for array implementation

```
void copyPaste(Stack src, int index)
```

which copies all the elements of the src stack and inserts to the stack at position index. The bottom element in the stack has index 0 as usual. You are not allowed to use pop, push, isEmpty functions. You can assume the destination stack has enough space for insertion. Your method should run in $\mathcal{O}(N)$ time.

For the original stack of

5
4
3
2

with parameter stack

7
8

with index = 1, the end result will be

5
4
3
7
8
2

For the original stack of

4
3
2

with parameter stack

6
7
8

with index = 0, the end result will be

4
3
2
6
7
8

IV. QUESTION (STACK) (20 POINTS)

Write the method

```
boolean checkString(int[] s)
```

which checks if the array s is of the form $1^m 2^n 3^n 4^m$ without counting any characters and using an external list based stack. You can assume the array consists of only numbers between 1 and 4, they are in that order, and there is at least one from each of them. You are only allowed to use pop, push, isEmpty functions. You should use a single external stack.

The method will return true for

1 1 1 2 2 3 3 4 4 4

and false for

1 1 1 2 2 2 2 2 3 3 3 3 3 4

The method will return true for

1 2 3 4

and false for

1 2 3 3 4